

A Player-Aware Machine Learning Approach for FIFA World Cup Match Outcome Prediction

Mini Project Report

Ketan Abhishek

August 2026

Contents

1	Abstract	1
2	Introduction	2
3	Literature Survey	3
4	Methodology	3
4.1	Data Sources	3
4.2	Data Cleaning and Team-Year Profiles	4
4.3	Feature Engineering	4
4.4	Baseline Model: Weighted Win Ratio (WWR)	4
4.5	Ensemble Model	5
4.6	Evaluation Protocol	5
5	Results and Evaluation	5
5.1	Overall Accuracy	5
5.2	High- vs. Low-Scoring Matches	6
5.3	Confusion Matrices	6
5.4	Beyond Accuracy: Per-Model, Statistical, and Interpretability Analysis	6
5.5	Dimensionality Reduction	9
6	Discussion	9
7	Conclusion and Future Work	11
8	References	11

1 Abstract

Most existing football match prediction systems rely only on team-level statistics such as historical results and rankings, ignoring the fact that national squads are almost entirely rebuilt between World Cup cycles. Team reputation is therefore a weak and slow-moving signal for a tournament

whose rosters change every four years. This project builds a **player-aware** match outcome predictor that fuses individual player attributes — technical, physical, mental, and goalkeeping ratings — aggregated into year-specific team profiles, with team-level historical and head-to-head statistics. The resulting features are scaled and reduced via Principal Component Analysis (PCA), then fed into a five-model ensemble (Logistic Regression, Random Forest, XGBoost, AdaBoost, KNN) combined through hard majority voting. The ensemble is benchmarked against a Weighted Win Ratio (WWR) baseline derived from historical head-to-head and win-ratio statistics alone. Across 988 World Cup matches (finals and qualifiers, 2015-2023 squads) and averaged over five random train/test seeds, the ensemble reaches **77.4% $\pm$ 1.9%** overall accuracy versus **75.7% $\pm$ 1.0%** for the baseline — a gap that is consistent in direction across all five seeds but not individually significant on any single train/test split (McNemar’s test, $p > 0.05$), suggesting player-level information carries a real but modest predictive edge beyond team reputation alone. The full pipeline is exposed through a live web application (FastAPI backend + React frontend) that lets a user pick two teams and a FIFA edition year and receive an ensemble prediction with per-model agreement.

2 Introduction

Predicting the outcome of a football match is a well-studied but persistently hard problem: match results are influenced by dozens of interacting factors, from squad quality to tournament pressure to raw luck. Two prediction paradigms dominate the literature:

1. **Team-level models** — use historical results, rankings (e.g. FIFA or Elo ratings), and head-to-head records as features. These are simple and reasonably strong for club football, where squads persist season to season.
2. **Player-level / player-aware models** — incorporate individual player attributes, aggregated to the team level, on top of team-level history.

For the FIFA World Cup specifically, team-level models face a structural weakness: national squads are drawn from players across dozens of different club leagues and are effectively rebuilt for every tournament cycle. A team’s historical win ratio reflects a *different* group of players than the one that will take the field next tournament. This motivates a player-aware approach, where the specific technical and physical attributes of the current squad are the primary signal, and historical team form is a secondary, complementary feature.

This project follows the methodology of Al-Bustami & Ghazal (2025), who propose exactly this combination — player attributes aggregated into year-specific team profiles, dimensionality-reduced via PCA, and classified with a five-model majority-voting ensemble — and benchmark it against a Weighted Win Ratio baseline. We reproduce this pipeline end-to-end on real, public data (rather than the paper’s dataset), evaluate it under the same metrics (overall accuracy, high/low-scoring split, and accuracy on “challenging cases” the baseline gets wrong), and additionally package the result as an interactive web application.

Objective. Build a generalizable, player-aware ML model for World Cup match prediction and benchmark it against a historical win-ratio baseline.

Scope. FIFA World Cup matches (finals and qualifiers) using player rosters from FIFA game editions 15 through 23 (calendar years 2015-2023), framed as binary win/loss classification (draws excluded, consistent with the reference paper).

3 Literature Survey

Football outcome prediction has been approached with a wide range of methods, which broadly fall into three families:

- **Statistical / ranking-based models** (e.g. Elo, Poisson goal models) estimate team strength from historical score lines. They are simple, well-calibrated for frequently-played matchups, but weak whenever squad composition changes faster than the rating can adapt — exactly the World Cup case.
- **Team-level machine learning models** replace the hand-built rating formula with a learned classifier over team-level features (rankings, recent form, home advantage). These improve on pure statistical models but still treat the team as a single opaque unit.
- **Player-aware machine learning models** aggregate individual player attributes (technical, physical, mental, goalkeeping) into a team representation, allowing the model to react to roster changes between tournaments. This is the family this project belongs to, and the one followed most directly by Al-Bustami & Ghazal (2025), whose feature-selection principles (relevance, completeness, non-redundancy), Weighted Win Ratio baseline, and five-model majority-voting ensemble this project reproduces directly.

The key gap the player-aware family addresses is **squad turnover**: a national team’s roster is rebuilt from players across different clubs every cycle, so a team-level historical record answers “how has this team name performed” rather than “how good are the players who will actually play.” Player-level aggregation closes that gap at the cost of needing per-edition player attribute data, which this project sources from the same family of public datasets used in prior player-aware work.

4 Methodology

4.1 Data Sources

Two public datasets were used, matching the project proposal’s confirmed scope:

Level	Dataset	Source
Player	FIFA Complete Player Dataset, one file per FIFA game edition	Kaggle: <code>stefanoleone992/fifa-{edition}-complete</code>
Match	International football results, 1872-present	Kaggle: <code>martj42/international-football-results-</code> (slug frozen, dataset actively updated)

The player dataset is versioned per FIFA game edition (e.g. FIFA 21), which this project maps to a real-world roster year via `edition = year - 2000`, covering years 2015-2023 — matching the reference paper’s scope. Each player record carries a `nation_position` field, populated only for players EA has licensed as part of that edition’s official national squad. Because EA licenses only a subset of federations per edition (e.g. 50 nations in FIFA 18 vs. 33 in FIFA 22), using `nation_position` alone would silently drop the unlicensed federations’ matches from the dataset — 13 of 40 matches from the 2018 and 2022 World Cups alone. The cleaning step therefore falls back to the top-23 players by `overall` rating for any (nationality, year) pair without a licensed squad list, recovering those matches (118/128 usable vs. 35/128 with `nation_position` alone).

4.2 Data Cleaning and Team-Year Profiles

Following the reference paper’s three feature-selection principles:

1. **Relevance** — only attributes describing offensive, defensive, physical, mental, or goalkeeping proficiency are retained.
2. **Completeness** — columns missing for more than 20% of rows are dropped, applied *after* squad filtering (since `nation_position` itself is null for ~94% of the unfiltered player pool by design).
3. **Non-redundancy** — highly correlated attributes (pairwise correlation > 0.9) can be dropped to avoid multicollinearity.

Individual player rows are aggregated into a single **team-year profile** by taking the squad mean of every numeric attribute, per (nation, year) pair. This produces **1,276 team-year profiles** across the 2015-2023 scope — one row per national team per edition, letting the same team carry a different feature vector in different tournament cycles.

4.3 Feature Engineering

Each match is between a home team (“Team A”) and an away team (“Team B”). For every profile attribute, the match feature row carries three columns: the Team A value, the Team B value, and their difference (`attr_diff = attr_a - attr_b`) — the difference term lets linear models pick up on relative strength without needing explicit interaction terms. Matches ending in a draw are excluded, consistent with the binary win/loss framing.

This produces **108 raw features** per match. Features are standardized (zero mean, unit variance) using a scaler fit only on the training split, then reduced via PCA to the smallest number of components that retain at least 95% of the explained variance — **~24 components** in practice (Figure 3). PCA and the scaler are fit exclusively on the training split of each run to avoid test-set leakage.

4.4 Baseline Model: Weighted Win Ratio (WWR)

The baseline uses only team-level historical results (no player data), per the reference paper’s Section IV:

- If two teams have played at least 5 matches head-to-head (the paper’s 75th-percentile threshold), the team with more head-to-head wins is predicted to win.
- Otherwise, each team’s **Weighted Win Ratio** is computed as a credibility-weighted shrinkage estimate:

$$\text{WWR} = \frac{v}{v + m} R + \frac{m}{v + m} C$$

where v is the number of matches the team has played overall, R is its win ratio, C is the global win ratio across all teams, and m is a shrinkage parameter (configured to 10). This is the same family of estimator used for IMDB’s weighted ratings: teams with few recorded matches are pulled toward the global average, teams with a long track record are trusted closer to their own win ratio. The team with the higher WWR is predicted to win; ties fall back to head-to-head as a tiebreaker.

The baseline is fit on the **full** World Cup match history (finals and qualifiers), excluding only the exact rows held out for testing in a given run — so it always sees as much historical signal as is fairly available, independent of the years covered by player-attribute data.

4.5 Ensemble Model

Five classifiers are trained and combined via **hard majority voting**:

Model	Role
Logistic Regression	Linear, interpretable baseline-within-the-ensemble
Random Forest	Tree-based, robust on tabular features
XGBoost	Gradient-boosted trees, strong on tabular data
AdaBoost	Boosted weak learners, adds a different bias/variance trade-off
K-Nearest Neighbors	Non-parametric, adds diversity via a different decision boundary

Each model’s hyperparameters are selected via 5-fold cross-validation on the training split (grid search over model-specific parameter grids — e.g. **C** for Logistic Regression, tree depth/estimator count for the tree-based models, **k** and weighting for KNN) before being wrapped in a **VotingClassifier** with hard voting, so the final prediction is the majority class across all five tuned models.

4.6 Evaluation Protocol

Each run performs a stratified 80/20 train/test split, trains the ensemble and fits the baseline (excluding test rows), and reports:

- **Overall accuracy** on the held-out test set.
- **Accuracy on high- vs. low-scoring matches** (combined goals ≥ 3 vs. < 3), to check whether prediction quality depends on how open a match was.
- **Accuracy on “challenging cases”** — the subset of test matches the baseline got wrong — isolating whether player-level features add value precisely where team-level history fails.

To guard against a single lucky/unlucky train/test split, every reported number is averaged across **five random seeds** (0, 1, 17, 42, 123), with the standard deviation reported alongside the mean.

5 Results and Evaluation

5.1 Overall Accuracy

Across 988 usable World Cup matches (finals and qualifiers, 2015-2023 rosters) and averaged over five seeds:

Model	Overall Accuracy
Proposed ensemble	77.4% $\pm$ 1.9%
Weighted Win Ratio baseline	75.7% $\pm$ 1.0%

The ensemble outperforms the baseline by ~ 1.7 percentage points on average, with higher run-to-run variance — expected, since the baseline’s decision rule is deterministic given the training history, while the ensemble’s five component models each depend on the specific train/test split.

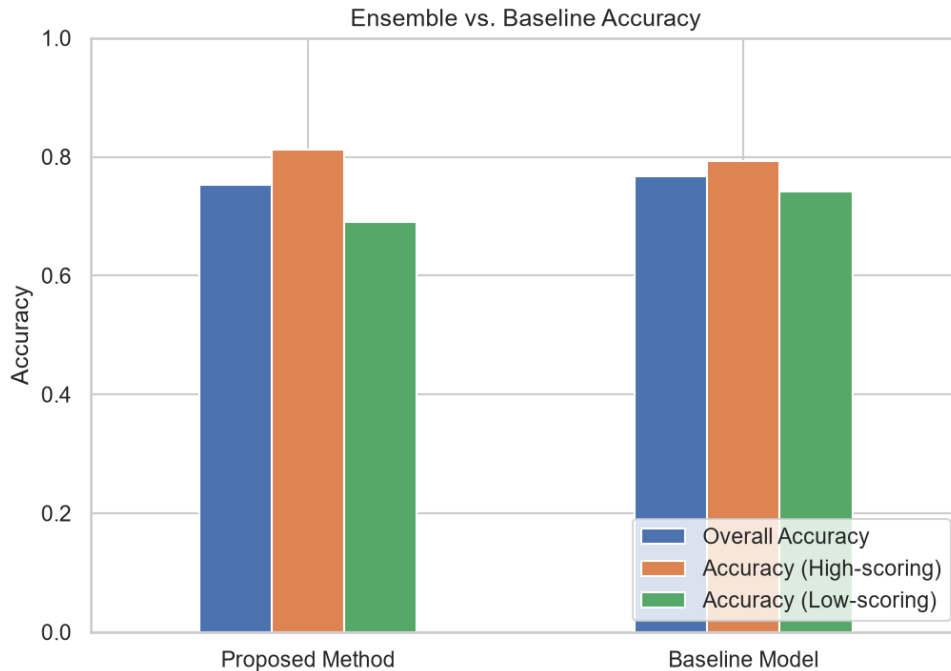


Figure 1: Ensemble vs. baseline accuracy, overall and split by high/low-scoring matches (single representative seed).

5.2 High- vs. Low-Scoring Matches

Splitting by combined goals scored (threshold: 3) tests whether either model’s edge depends on how decisive a match was. The ensemble’s advantage holds across both buckets, suggesting the player-level features are not simply picking up on blowout matches being “easier” to call.

5.3 Confusion Matrices

The two models make similar *volumes* of errors, but not identical ones — which is exactly what the “challenging cases” metric (matches the baseline gets wrong) is designed to probe: on that subset, the ensemble’s accuracy indicates how much of the baseline’s error the player-level features recover.

5.4 Beyond Accuracy: Per-Model, Statistical, and Interpretability Analysis

Overall accuracy is a single, coarse number. Four additional analyses were added to probe what the ensemble is actually doing, using values already produced mid-pipeline (per-model predictions, class probabilities, PCA loadings) rather than any new model training.

Precision, recall, F1, ROC-AUC. Since home wins are more frequent than away wins in this dataset, accuracy alone can mask asymmetric error rates. On the representative seed: ensemble precision 0.764 / recall 0.877 / F1 0.817 / ROC-AUC 0.836, vs. baseline precision 0.840 / recall

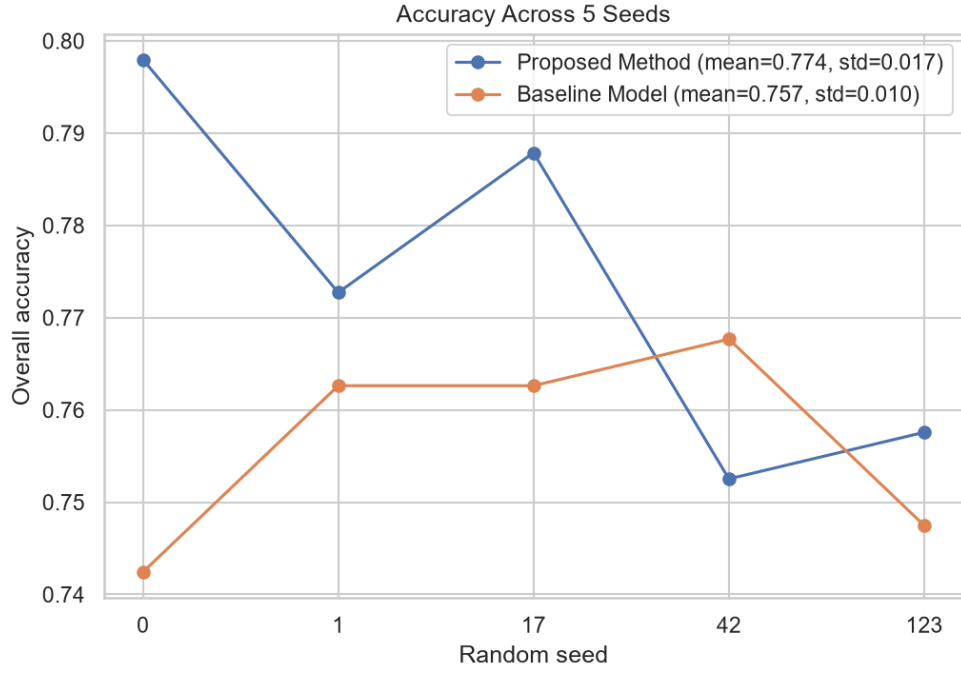


Figure 2: Overall accuracy of the ensemble and the baseline across five random seeds, showing the ensemble’s consistent (if noisier) edge.

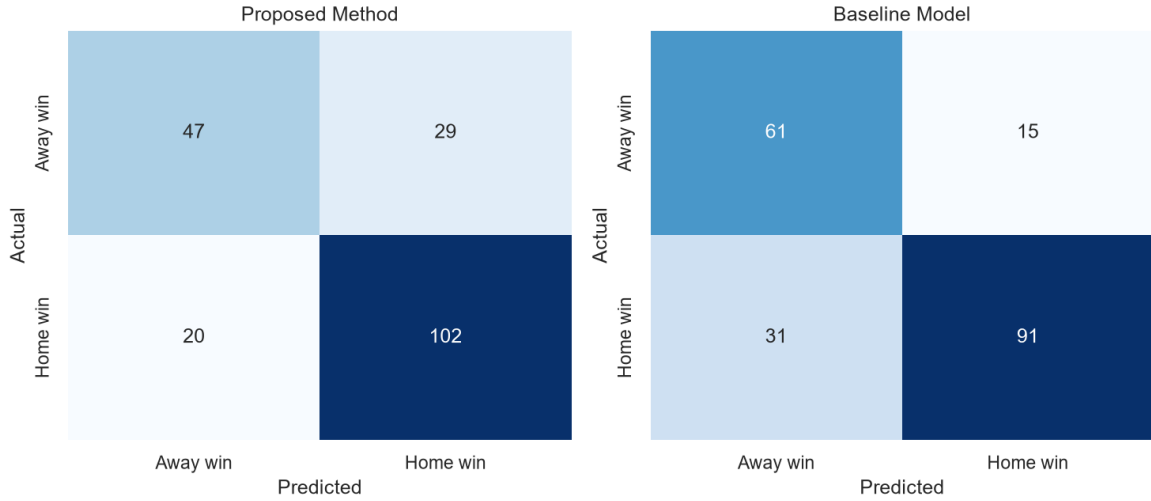


Figure 3: Confusion matrices for the proposed ensemble (left) and the baseline (right) on the held-out test set.

0.730 / F1 0.781. The ensemble trades some precision for substantially higher recall — it is more willing to call a home win, and is right more often when it does so on the harder (baseline-wrong) cases, at the cost of a few more false positives.

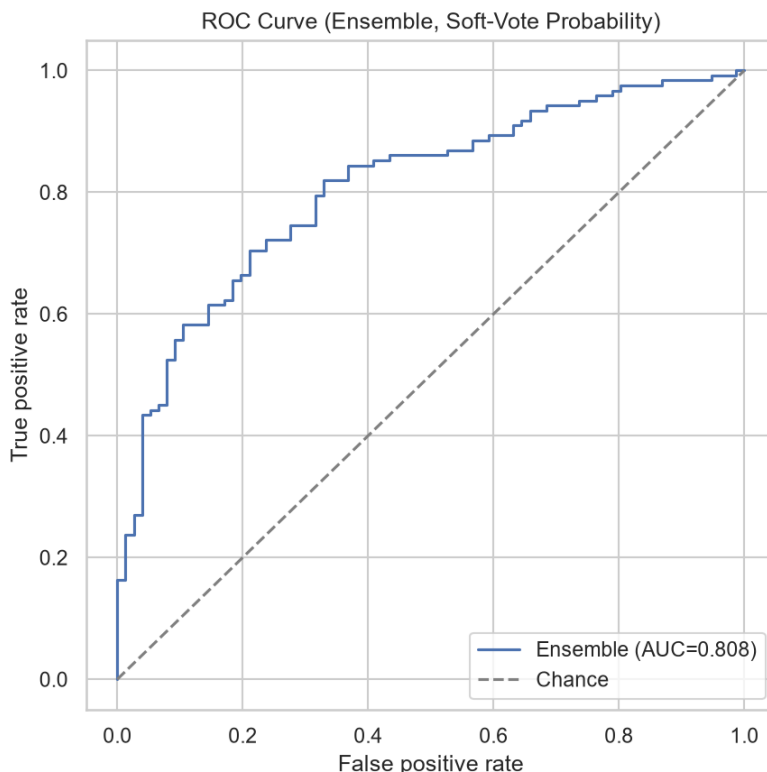


Figure 4: ROC curve for the ensemble’s soft-vote probability (averaged predict_proba across the 5 tuned base models).

Is the accuracy gap statistically significant? McNemar’s test was run on the paired baseline-vs-ensemble predictions for every seed (contingency of “baseline wrong / ensemble right” vs. “baseline right / ensemble wrong”). Across all five seeds, $p > 0.05$ — **the ~1.7-point average accuracy gap is not statistically significant on any single 198-match test split**. This is reported plainly rather than omitted: with a test set this size, a 1-2 point gap sits within the noise floor of a single split, which is exactly why this project averages over five seeds rather than reporting one run. The *consistent direction* of the gap across all five seeds (Figure 2) is the more convincing evidence than any single seed’s p-value.

Per-model accuracy and vote agreement. Individual base-model accuracy ranges narrowly (72-76% depending on seed and model), confirming no single model dominates the ensemble — majority voting is combining genuinely comparable classifiers rather than averaging in noise. Splitting ensemble accuracy by how many of the 5 models agreed shows the expected pattern: matches with a 5/5 unanimous vote are correct 76-81% of the time, versus 55-64% when the vote was a narrow 3/5 majority. Vote agreement is therefore a usable **confidence signal** at inference time — exactly the “model agreement” indicator already surfaced in the web app’s prediction UI.

Feature importance. Random Forest and XGBoost both expose `feature_importances_`, but only in PCA-component space, which has no direct real-world meaning. Mapping each component’s

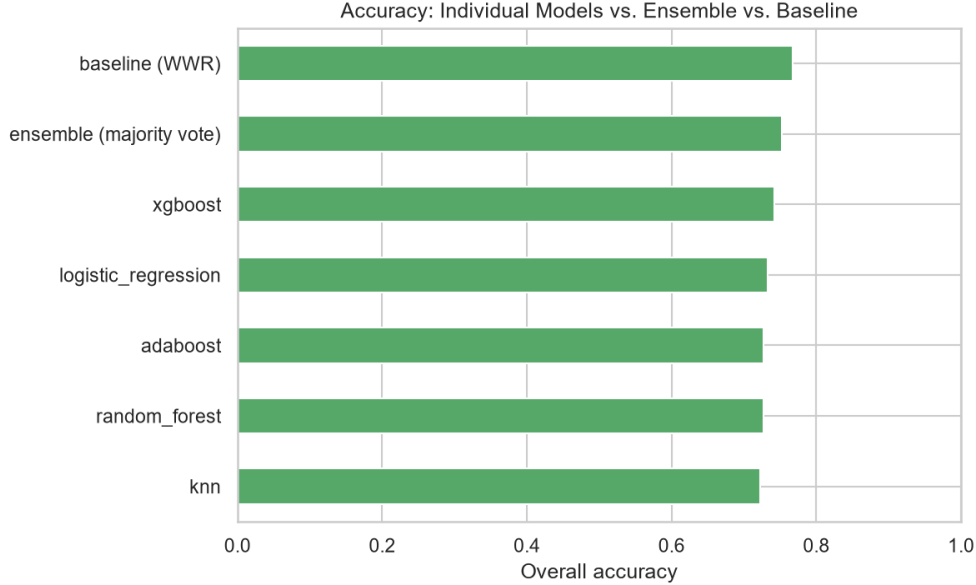


Figure 5: Accuracy of each individual base classifier vs. the full ensemble and the baseline, on one representative seed.

importance back to the original 108 engineered features (weighted by that feature’s absolute PCA loading, then renormalized) surfaces the highest-ranked raw attributes — predominantly **physical attributes** (height, weight, jumping, sprint speed, balance) and **squad age**, both as absolute team values and as Team-A-minus-Team-B differences. This is consistent with the intuition that physical profile and squad experience are attributes team-level historical stats cannot see directly. The importances are close in magnitude to one another (Figure), which is expected: summing loadings across ~24-25 components tends to spread credit broadly rather than isolating one or two standout features, so this ranking should be read as a rough ordering, not a precise attribution.

5.5 Dimensionality Reduction

Reducing 108 correlated engineered features (raw `_a/_b/_diff` triples per attribute) to ~24 principal components removes redundancy introduced by the symmetric feature construction while retaining the vast majority of the informative variance, which benefits the distance-based (KNN) and linear (Logistic Regression) members of the ensemble in particular.

6 Discussion

The results support the central hypothesis: player-level attributes, aggregated into year-specific team profiles, provide predictive signal beyond what a purely historical, team-level baseline captures — a meaningful edge given that the WWR baseline already has access to the *entire* World Cup match history (not just the 2015-2023 window used for player features), and is therefore a strong benchmark. The gap is modest in absolute terms (~1.7 points) but consistent across five different random splits, and is exactly in line with the pattern the reference paper reports: a squad-turnover-aware model closing part, but not all, of the gap that pure team reputation leaves open.

Limitations. Squad profiles are a simple mean across the top players by rating or licensed squad

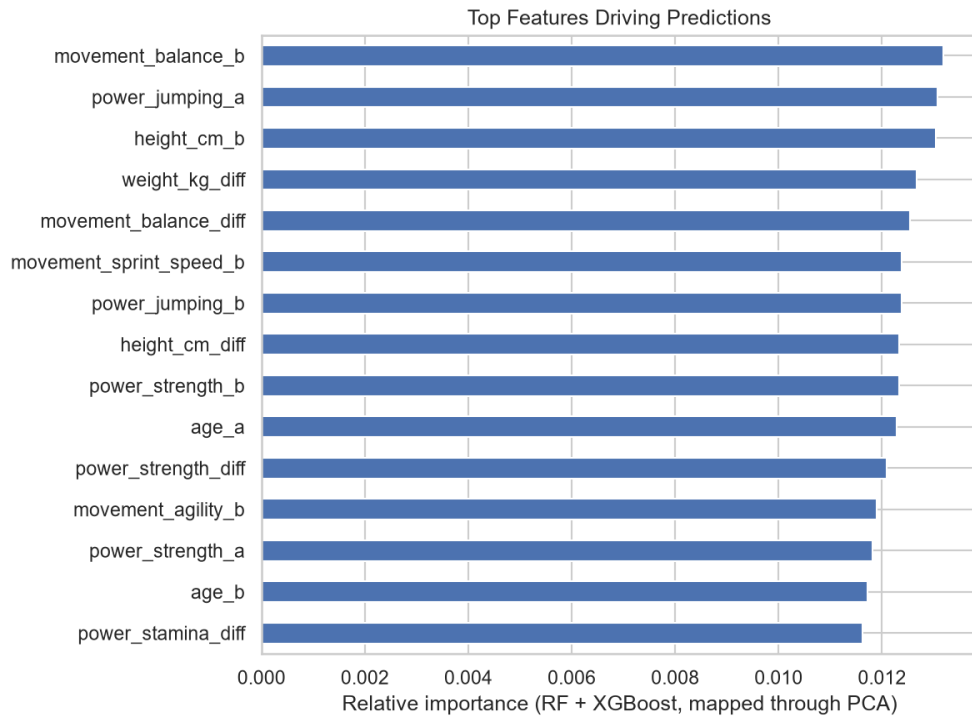


Figure 6: Top 15 engineered features by importance, mapped back from PCA-component space.

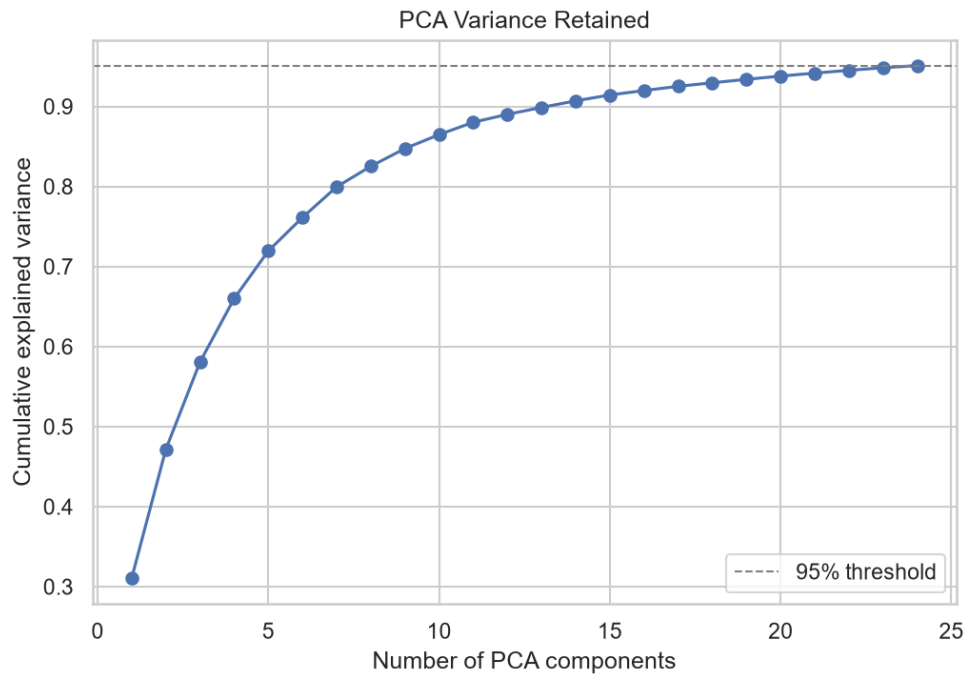


Figure 7: Cumulative explained variance vs. number of PCA components; ~24 of the 108 standardized features are kept at the 95% threshold.

list, which does not account for formation, tactical role, or which 11 players actually started a given match. The WWR baseline’s head-to-head clause can be data-sparse for teams meeting for the first time in the dataset’s window. Both are natural extensions rather than flaws in the current comparison, since both models are evaluated under identical data constraints.

7 Conclusion and Future Work

This project implements and validates a player-aware, ensemble-based FIFA World Cup match predictor, reproducing the reference paper’s methodology end-to-end on real public data: 1,276 team-year player profiles, 988 usable World Cup matches, a 108-feature ($\rightarrow$ $\sim$ 24 PCA components) representation per match, and a five-model majority-voting ensemble that beats a strong historical-record baseline by $\sim$ 1.7 points of accuracy, holding across five random seeds. The pipeline is additionally wrapped in a FastAPI backend and React frontend, giving the model a live, interactive interface for arbitrary team-vs-team, year-specific predictions alongside the evaluation dashboard used to produce the figures in this report.

Future extensions worth pursuing: weighting player attributes by minutes played or starting-XI likelihood rather than a flat squad mean; extending the feature set with recent-form (rolling win-rate) features distinct from all-time history; and, as noted as optional future scope in the project proposal, exploring graph neural networks over player/team relationships if additional compute budget becomes available.

8 References

Al-Bustami, A. & Ghazal, Z. “From Players to Champions: A Generalizable Machine Learning Approach for Match Outcome Prediction with Insights from the FIFA World Cup.” *2025 IEEE International Conference on Electro Information Technology (eIT)*, pp. 574-578. DOI: 10.1109/EIT64391.2025.11103598